The Crag Draw Network: Extended Anchor Equation Evidence for Cosmological Substrate Pre-Strain in the Burdick Crag Mass Framework

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Abstract

The Burdick Crag Mass (BCM) framework treats galactic rotation curves as evidence of a substrate field maintained by supermassive black hole (SMBH) neutrino flux injection. Here we extend BCM to a cosmological network model in which galaxies are classified as ROOT, BRANCH, LEAF, or VOID-EDGE crag nodes in a hierarchical substrate draw network. Applying this classification to 175 galaxies from the SPARC rotation curve catalog, we find that the BCM rotation curve correction is concentrated in MID-mass BRANCH-tier recipient galaxies (82.1% beat Newton) rather than HIGH-mass ROOT source galaxies (44.7%), consistent with ROOT nodes exporting substrate outward through the draw network. After removing 73 physically implausible MOND comparison rows, BCM advantage over MOND decreases with estimated SMBH mass proxy (rho = -0.42, N=102), inconsistent with local engine funding and consistent with cosmologically pre-strained substrate. Analysis of 21 galaxies using real Planck 2018 SMICA CMB temperature data reveals that Planck thermal and CMB-frame velocity proxies for primordial alignment are orthogonal (rho = -0.047), indicating two independent signals for the substrate topology. An empirically locked unified alignment signal (60% Planck thermal, 40% kinematic) identifies nine stable backbone crag nodes in the Local Volume. Together these results support the Primordial Gutter Hypothesis: the BCM substrate field was initialized by a Bang-type rip at universe formation, maintained locally by SMBH activity but not generated by local engines.

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1. Introduction

The missing mass problem in galactic rotation curves has been addressed through dark matter halo models (Rubin et al. 1980; Navarro, Frenk & White 1997) and Modified Newtonian Dynamics (Milgrom 1983; Famaey & McGaugh 2012). The Burdick Crag Mass (BCM) framework (Burdick 2025a, 2025b) proposes a third mechanism: a substrate field sigma maintained by SMBH neutrino flux injection (J-Vorticity) that provides additional centripetal support at large galactic radii.

BCM was validated against 175 galaxies from the Spitzer Photometry and Accurate Rotation Curves (SPARC) catalog (Lelli, McGaugh & Schombert 2016) with zero free parameters. A five-term Extended Anchor Equation governs the substrate field, with Terms 1 (Einstein recovery) and 5 (entropy sink) at highest empirical confidence and Term 3 (spectral projection gate R_9to10) confirmed velocity-independent across all tested regimes (Burdick 2025b, Test 18).

In this work we report three new findings from BCM v28 (Burdick 2026): (i) The BCM rotation curve correction signal has a network topology fingerprint concentrated in intermediate-mass BRANCH recipient galaxies, not massive ROOT source galaxies. (ii) After MOND sanity filtering, BCM advantage over MOND decreases with SMBH mass proxy, inconsistent with local engine funding. (iii) Real Planck 2018 CMB data reveals that thermal and kinematic alignment proxies are orthogonal, pointing to a cosmological origin for the substrate field.

2. Crag Network Classification

2.1 Crag Intensity Index

We define the Crag Intensity Index for each galaxy as C_I = J_amp x sigma_deficit, where J_amp = $(V\_max / V\_ref)^2$ x J_ref is the substrate current injection amplitude normalized to the Test 13 reference galaxy NGC 5055 (V_ref = 206 km/s, J_ref = 8.0), and sigma_deficit = sigma_crit x (J_amp / J_ref) x N is the total substrate displaced per pierce (sigma_crit = 5×10^{-4}, N = 60 steps). Galaxies are classified into four tiers:

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ROOT : C_I > 0.1 (N = 53 in SPARC 175)
BRANCH : 0.01 < C_I <= 0.1 (N = 39)
LEAF : 0.001 < C_I <= 0.01 (N = 58)
VOID-EDGE : C_I <= 0.001 (N = 25)
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2.2 The Dual-Flow Model

Applying a J kill chain (substrate pump set to zero) across all 175 galaxies, we measure substrate funding fraction — the fraction of rotation curve improvement attributable to substrate vs Newton. The tier-mean fractions are:

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ROOT: 0.111 BRANCH: 0.433
LEAF: 0.459 VOID-EDGE: 0.276
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The non-monotonic gradient (BRANCH > LEAF > VOID > ROOT) is the expected signature of a draw network in which ROOT galaxies are pump sources that export substrate outward. ROOT systems have high intrinsic battery (mean 0.889) and low own-consumption because the substrate they generate flows to BRANCH and LEAF recipients. The cascade propagation model predicts a delay t_i = t_ROOT + d_i / C_network before recipient nodes feel the cascade of a ROOT pump failing, where d_i is the network distance (Tests 24–25).

3. Rotation Curve Evidence

3.1 BCM vs Newtonian Dynamics

Across all 175 SPARC galaxies, BCM beats the Newtonian prediction on 122/175 (70%) with zero free parameters. The win rate stratified by V_max group is:

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HIGH-mass (V_max > 200 km/s, ROOT tier): 44.7%
MID-mass (80 <= V_max <= 200 km/s): 82.1%
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LOW-mass (V_max < 80 km/s, VOID-EDGE tier): 78.0%
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This inversion — BCM performs worst against Newton on the most massive galaxies — is the rotation curve fingerprint of the dual-flow model. ROOT galaxies export substrate; their own rotation curves show the smallest BCM correction above Newton. MID-mass BRANCH recipients show the largest correction.

3.2 BCM vs MOND (sanity-filtered)

Comparing BCM against MOND (a_0 = 1.2 x 10^{-10} m/s^2, standard interpolating function), we first apply five sanity gates to detect impossible comparison rows: BCM advantage exceeding 75% of peak rotation velocity, non-finite RMS values, or BCM advantage >30 km/s in dwarfs (V_max < 80 km/s). These gates flag 73/175 (42%) rows. Gate 2 (finite/nonneg RMS) flags zero rows, confirming the implementation runs without numerical failure.

After removing flagged rows (Test 29), the rank correlation between estimated SMBH mass proxy (log M_BH from M-sigma via sigma = 0.65 x V_max, Kormendy & Ho 2013) and BCM fractional advantage over MOND is:

rho(log M_BH, BCM/MOND fraction) = -0.42 (N = 102 clean rows)

Larger estimated engine mass yields smaller fractional BCM advantage over MOND. The low-mass anomaly is particularly informative: 59/59 LOW-mass galaxies (V_max < 80 km/s, estimated M_BH < 10^6 M_{sun}) show BCM beating MOND, with the largest fractional advantages. Galaxies with negligible SMBHs show the strongest relative BCM signal, ruling out local engine generation as the dominant mechanism.

4. CMB Alignment Analysis

4.1 Two independent alignment proxies

For 21 galaxies with CMB-frame systemic velocities (V_3K from NED/literature), we construct two independent proxies for primordial substrate alignment:

A_CMB_planck = tanh(delta_T_uK / 70)

where delta_T is the Planck 2018 SMICA temperature at the galaxy sky position (real pixel values at nside=64, BCM v28 Test 22), and

A_CMB_v3k = -tanh(V_peculiar / 500)

where V_peculiar = V_3K - H_0 x d (H_0 = 70 km/s/Mpc). The rank correlation between A_CMB_planck and A_CMB_v3k is rho = -0.047. The two proxies are orthogonal — they measure independent physical quantities. The Planck proxy measures primordial thermal scar topology. The kinematic proxy measures current substrate pressure. Neither is a proxy for the other.

4.2 Unified alignment signal

A weight sweep across 11 combinations (w_t = 0 to 1) shows that rank correlation with the base crag intensity C_I is maximized and tier classification flips are minimized at the same weight:

A_CMB_full = 0.60 x A_planck + 0.40 x A_v3k (LOCKED)

At this weighting: rank correlation rho = 0.895, minimum 6 tier flips across 21 galaxies (Test 23). Nine of 21 galaxies are classification-stable across the full weight sweep — confirmed backbone nodes of the Local Volume crag network.

4.3 Crag phase states

Combining C_I, A_CMB_full, substrate fraction proxy, and V_peculiar, galaxies are classified into phase states (Test 27): ACTIVE_SHEAR_CRAG (ROOT tier, V_pec > 0; N=9), RETURNING_ROOT_CRAG (ROOT tier, V_pec < 0; N=3), DEPENDENT_BRANCH (BRANCH/LEAF tier; N=6), CLUSTER_CONTAMINATED (N=2). The local volume is predominantly in active shear phase — the Bang rip is still propagating outward in most surveyed galaxies.

5. Discussion

5.1 Separation of three quantities

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Crag tier Substrate capacity / network depth
SMBH mass Local pump regulator, not generator
Substrate fraction Current dependency on network flow
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ROOT galaxies host the largest estimated SMBHs but show the lowest substrate dependency (0.111). This inversion is consistent with ROOT nodes being pump-and-draw junctions that distribute substrate outward while drawing from surrounding void — not consuming it locally.

5.2 The Primordial Gutter Hypothesis

The low-mass anomaly (59/59 dwarf galaxies show BCM signal without meaningful SMBHs) requires a substrate source independent of the local engine. The Primordial Gutter Hypothesis (Burdick 2026, FIG. 13) proposes that the substrate field was initialized at universe formation as a simultaneous multithreaded manifold rip — a Bang-type rip — that produced a pre-strained substrate topology encoded in the CMB anisotropy field. The orthogonality of Planck thermal and kinematic CMB proxies (rho = -0.047) is consistent with these measuring independent aspects of this primordial structure.

The BCM Extended Anchor Equation Term 5 (entropy sink) provides the maintenance cost — without continuous SMBH funding, the substrate decays at rate lambda = 0.1 (normalized). The J kill chain confirms galaxies retain approximately 85% intrinsic support for ~19 BCM time units after pump failure, before the cascade reaches BRANCH and LEAF recipients.

5.3 Limitations

The SMBH mass proxy (M_BH from M-sigma via V_max, +/-0.5 dex) introduces substantial uncertainty. Cross-referencing with direct sigma or M_BH measurements from Greene et al. (2020) is required before the coupling test (Test 28) can be considered definitive. The network distance proxy (log-space C_I ratio) requires replacement with real 3D Mpc separations from Cosmicflows-4. The MOND comparison should be validated against published SPARC MOND residuals (Lelli et al. 2017) before the 42% flag rate is attributed solely to implementation error.

6. Conclusions

1. BCM rotation curve correction is concentrated in MID-mass BRANCH recipient galaxies (82.1% beat Newton) vs HIGH-mass ROOT source galaxies (44.7%), consistent with a substrate draw network in which ROOT nodes export substrate outward rather than consuming it locally.

2. After MOND sanity filtering (73/175 rows removed), BCM advantage over MOND decreases with SMBH mass proxy (rho = -0.42, N=102), inconsistent with local engine funding and consistent with cosmological substrate pre-strain.

3. Planck thermal and V_3K kinematic CMB alignment proxies are orthogonal (rho = -0.047), indicating two independent signals for primordial substrate topology. Optimal unified weighting: 60% Planck thermal / 40% kinematic (empirically locked, Test 23).

4. Nine stable backbone crag nodes are identified in the Local Volume whose classification is proxy-independent, forming the framework for a full 3D substrate draw field map.

5. These results support the Primordial Gutter Hypothesis: the BCM substrate field is cosmologically pre-strained, maintained but not generated by local SMBH activity.

Data Availability

All data, code, and test results are available at Zenodo DOI 10.5281/zenodo.20120354 (BCM v28) and GitHub: https://github.com/Joy4joy4all/Burdick-Crag-Mass. BCM substrate solver, test scripts (Tests 17–30), rotation curve fits, Planck pixel extraction utility (BCM_v28_EXTRACT_PLANCK_PIXELS.py), and bcm_planck_galaxy_pixels.json are included.

References

Burdick S.J. 2025a, Zenodo, doi:10.5281/zenodo.19251192 (BCM v1–v24)

Burdick S.J. 2025b, Zenodo, doi:10.5281/zenodo.20038700 (BCM v27)

Burdick S.J. 2026, Zenodo, doi:10.5281/zenodo.20120354 (BCM v28, this work)

Famaey B., McGaugh S.S. 2012, Living Reviews in Relativity, 15, 10

Gorski K.M. et al. 2005, ApJ, 622, 759 (HEALPix) doi:10.1086/427976

Greene J.E. et al. 2020, ARA&A, 58, 257

Kormendy J., Ho L.C. 2013, ARA&A, 51, 511

Lelli F., McGaugh S.S., Schombert J.M. 2016, AJ, 152, 157 (SPARC)

Lelli F. et al. 2017, ApJ, 836, 152

Milgrom M. 1983, ApJ, 270, 365

Navarro J.F., Frenk C.S., White S.D.M. 1997, ApJ, 490, 493

Planck Collaboration (Aghanim et al.) 2020, A&A, 641, A1 doi:10.1051/0004-6361/201833880

Planck Collaboration (Akrami et al.) 2020, A&A, 641, A4 (SMICA) doi:10.1051/0004-6361/201833881

Rubin V.C., Ford W.K., Thonnard N. 1980, ApJ, 238, 471

Tremaine S. et al. 2002, ApJ, 574, 740

Tully R.B., Fisher J.R. 1977, A&A, 54, 661

Zonca A. et al. 2019, JOSS, 4(35), 1298 (healpy) doi:10.21105/joss.01298

"The substrate is inherited from the Bang, not generated by the current central engine."
— Stephen Justin Burdick Sr., 2026

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